HOME AUTOMATION SYSTEM

Robotics & AI

IA 1 REPORT

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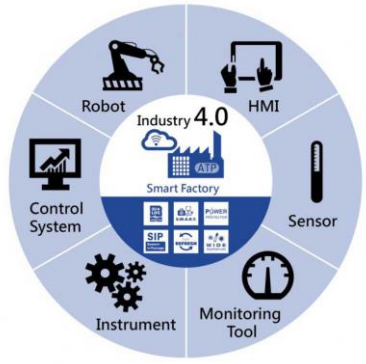
Contents

1	Abstract
2	Introduction
3	Circuitry
4	Working
5	Applications
6	Future Scope
7	Conclusion
8	References

Abstract

We are now in the fourth industrial revolution, also referred to as Industry 4.0. It is characterized by increasing automation and the employment of smart machines and smart factories, informed data helps to produce goods more efficiently and productively across the value chain. Flexibility is improved so that manufacturers can better meet customer demands using mass customization—ultimately seeking to achieve efficiency with, in many cases, a lot size of one. By collecting more data from the factory floor and combining that with other enterprise operational data, a smart factory can achieve information transparency and better decisions.

The term IoT, or Internet of Things, refers to the collective network of connected devices and the technology that facilitates communication between devices and the cloud, as well as between the devices themselves. Thanks to the advent of inexpensive computer chips and high bandwidth telecommunication, we now have billions of devices connected to the internet. This means everyday devices like toothbrushes, vacuums, cars, and machines can use sensors to collect data and respond intelligently to users. The Internet of Things integrates everyday “things” with the internet. The cost of integrating computing power into small objects has now dropped considerably. A whole industry has sprung up with a focus on filling our homes, businesses, and offices with IoT devices. These smart objects can automatically transmit data to and from the Internet. All these “invisible computing devices” and the technology associated with them are collectively referred to as the Internet of Things.



Introduction

Home automation is a network of hardware, communication, and electronic interfaces that work to integrate everyday devices with one another via the Internet. Each device has sensors and is connected through WiFi, so you can manage them from your smartphone or tablet whether you're at home, or miles away. This allows you to turn on the lights, lock the front door, or even turn down the heat, no matter where you are. There are three main elements of a home automation system: sensors, controllers, and actuators.

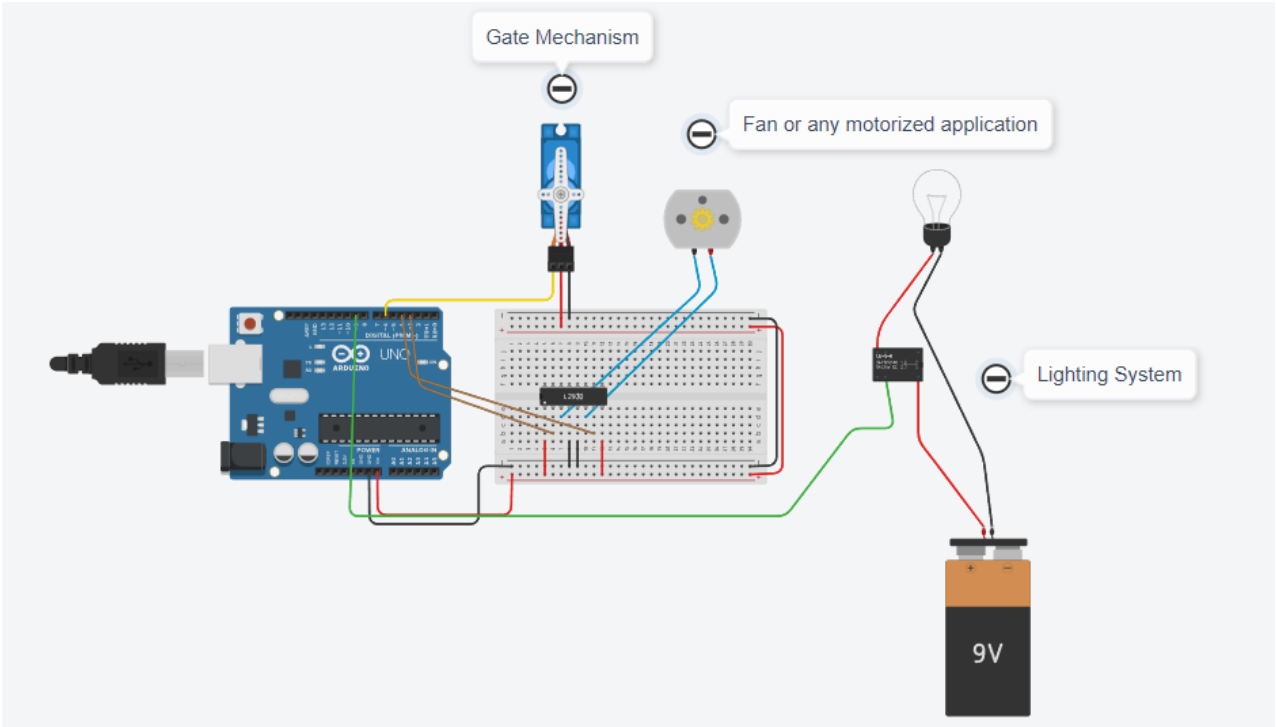
- **Sensors** can monitor changes in daylight, temperature, or motion detection. Home automation systems can then adjust those settings (and more) to your preferences.
- **Controllers** refer to the devices, personal computers, tablets or smartphones, used to send and receive messages about the status of automated features in your home.
- **Actuators** may be light switches, motors, or motorized valves that control the actual mechanism, or function, of a home automation system. They are programmed to be activated by a remote command from a controller.

The purpose of a home automation system is to streamline how your home functions. In this report, we present the home automation system developed by us, using various devices. We have preferred a cost-effective approach for our project, enabling us to develop this system and showcase its working with utmost commitment. The objective here is to learn from our findings and use our knowledge to work on this project to deliver what we have intended from the beginning.

Home Automation offers a variety of benefits:

- **Remote access:** Control your home from mobile devices, including your laptop, tablet, or smartphone.
- **Comfort:** Use home automation to make your home a more comfortable, livable space. Preprogram your thermostat with your preferred settings so that your home is always at a comfortable temperature, set up smart speakers to play music when you get home from work, or adjust your lights to soften or brighten based on the time of day.
- **Convenience:** Program devices to turn on automatically at certain times, or access their settings remotely from anywhere with an Internet connection. When you don't have to remember to lock the door behind you or switch off the lights, you can turn your attention to more important things.
- **Increased safety:** Smart fire detectors, carbon monoxide monitors, pressure sensors, and other home automation security features can help protect your home from disaster.
- **Energy efficiency:** Home automation allows you to be more mindful of your power usage. For example, you can save on energy bills by reducing the length of time that lights stay on, or by lowering temperatures when you leave a room.

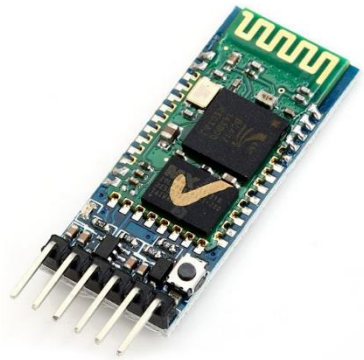
Circuitry

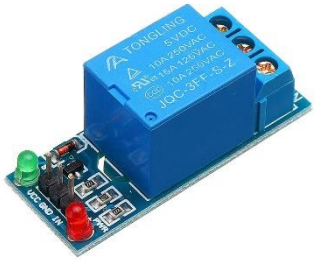


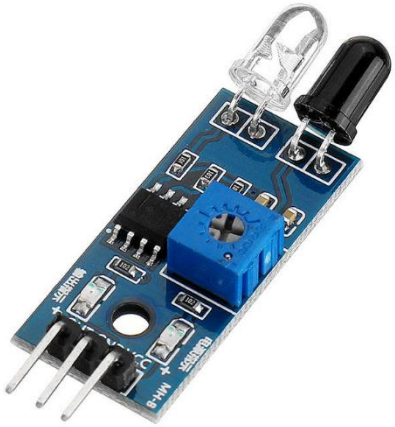
The system is an Arduino project. It makes use of Arduino based devices. An actual electric bulb has been used in the system as it achieves the objective of what home automation can do. The parts used in the system are as follows:

- **Arduino UNO Board:** Arduino Uno is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button.
- **Bread Board:** A solder less bread board to establish connections between different devices.
- **Bluetooth Module:** A Bluetooth module is used to operate the devices from an application in smart phone. The model used is called HC 05.

- **Relay:** It has been added to use an electric bulb in the circuit. From a relay module we can connect a 230V supply with our max. 5V operated circuit.
- **Electric Bulb:** It is one of the applications of home automation. We operate it through our phones.
- **DC Motor:** We make use of a dc motor to operate a small fan. We control it from our smart phones.
- **Motor Driver:** A motor driver has been used to connect the dc motor with the Arduino board. The IC used is called L293D.
- **IR Sensor:** An IR sensor senses and obstruction in the infrared light path. It senses the signal and actuates the gate mechanism.
- **Servo Motor:** A servo motor is used as a gate mechanism for the house. The house gate opens and closes with the help of the IR sensor described above.







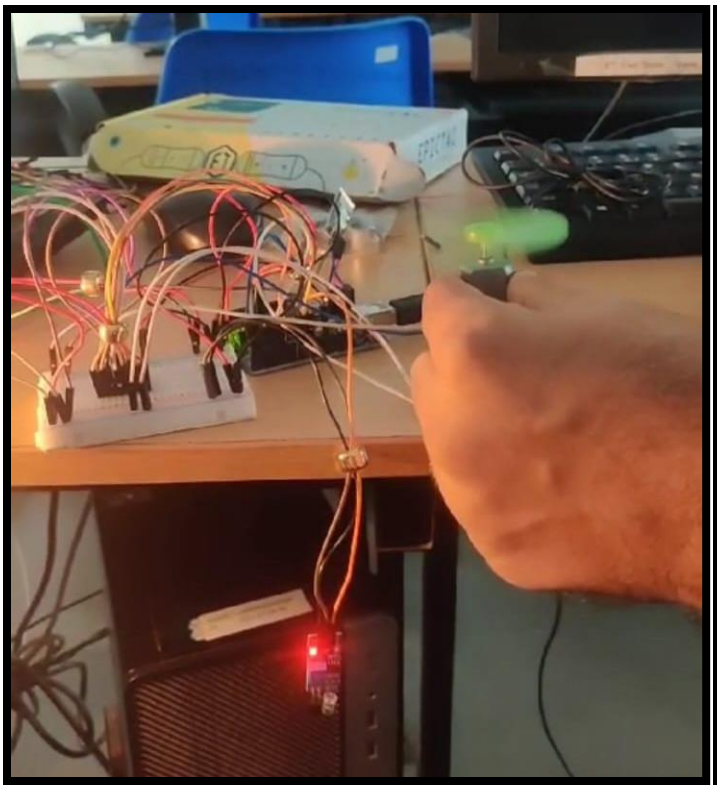
Working

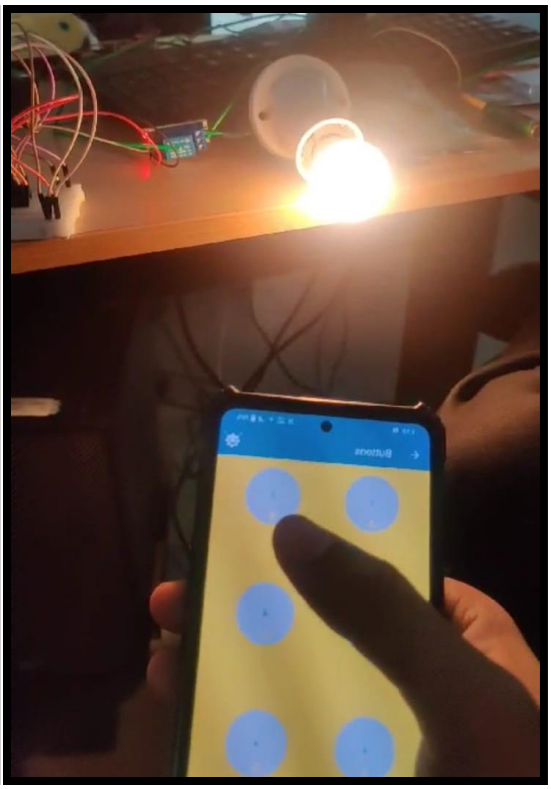
The system operates on Bluetooth and IR sensor. An application called “Arduino Bluetooth Control” should be installed on your smart phone. When your device is paired with the Bluetooth, you can switch ON and OFF the bulb as well as the motor from the respective buttons present in the application. The IR sensor controls the servo motor. When it senses an obstacle in its path of light ray, the servo motor actuates and the gate opens. The reverse of this takes place when the obstacle is removed. The entire setup has been programmed on the Arduino IDE software. The code for the same is as follows:

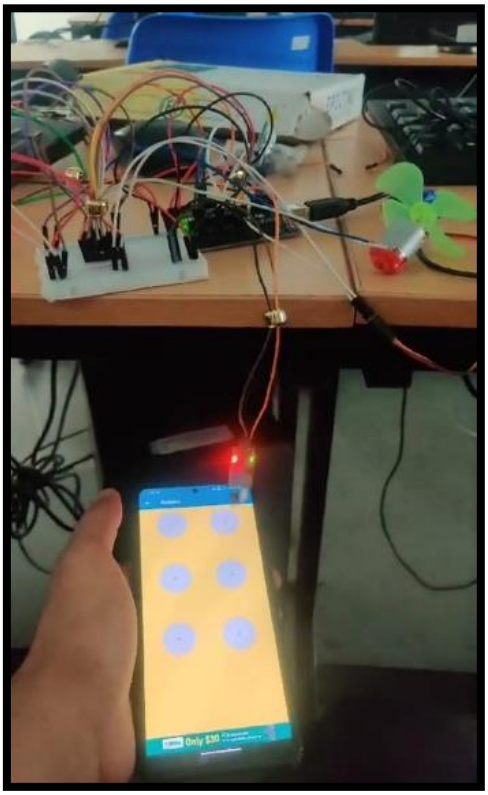
```
#include <Servo.h>
int bulb= 13; //Bulb
char val;
int Motor1a = 8; //Motor Fan
int Motor1b = 9;
Servo myservo;
int IR = 10;
void setup()
{
pinMode(bulb,OUTPUT);
pinMode(Motor1a,OUTPUT);
pinMode(Motor1b,OUTPUT);
myservo.attach(4);
Serial.begin(9600);
digitalWrite(13,LOW);
}

void loop()
{
int data = digitalRead(IR);
{
   if(data==LOW)
    {
    myservo.write(90);
```

```
    delay(15);
    }
    else
    {
    myservo.write(0);
    delay(15);
    }
}
if(Serial.available())
{
 val = Serial.read();
 Serial.println(val);
}
if(val=='1') //Light bulb ON
{
  digitalWrite(13,HIGH);
}
else if(val=='2') //Light bulb OFF
{
  digitalWrite(13,LOW);
}
else if(val == '3') //Motor ON
{
  digitalWrite(8,HIGH);
}
else if(val == '4') //Motor OFF
{
  digitalWrite(8,LOW);
}
}
```

Photos







Applications

There are numerous applications of home automation, which can enhance convenience, energy efficiency, security, and overall quality of life. Here are some common applications:

- **Lighting Control:** Homeowners can remotely control and automate their lighting systems. This can include adjusting brightness, scheduling lights to turn on/off at specific times, or using motion sensors to activate lights when someone enters a room. This can save energy and enhance security.
- **Thermostat Control:** Smart thermostats allow for precise control of heating and cooling systems. They can learn your preferences, adjust temperature based on occupancy, and provide energy savings by optimizing heating and cooling cycles.
- **Security:** Home automation systems can integrate security cameras, motion detectors, and door/window sensors. Users can monitor their property remotely, receive alerts, and even automate actions like locking doors or turning on lights when motion is detected.
- **Entertainment:** Automation can be applied to home entertainment systems. This includes controlling TVs, audio systems, and streaming devices through a single interface or using voice commands.
- **Appliance Control:** Smart plugs and outlets can be used to control appliances like coffee makers, irons, or even ovens remotely. This can be convenient and energy-saving.
- **Smart Locks:** Automated door locks enable remote locking/unlocking and can grant access to specific individuals with temporary digital keys. This is useful for home security and granting access to guests or service providers.
- **Voice Assistants:** Integrating voice-activated assistants like Amazon Alexa or Google Assistant can control various aspects of the home, from adjusting the thermostat to playing music.

- **Energy Efficiency:** Home automation can help save energy by optimizing heating and cooling, turning off lights and appliances when not in use, and providing real-time energy consumption data.
- **Automated Window Coverings:** Motorized blinds or shades can be scheduled to open and close at specific times to control natural light, temperature, and privacy.
- **Irrigation and Garden Control:** Smart irrigation systems can adjust watering schedules based on weather conditions, soil moisture levels, and plant types, conserving water and maintaining a healthy garden.
- **Health and Wellbeing:** Home automation can support health-related applications, such as monitoring air quality, controlling air purifiers, or even reminding individuals to take medication.
- **Home Theater and Gaming Rooms:** Customized lighting, audio, and video setups can be automated for an immersive entertainment experience.
- **Smart Kitchen:** Appliances like smart refrigerators, ovens, and coffee makers can be controlled remotely, provide recipe suggestions, and track food inventory.
- **Aging in Place:** Home automation can be used to assist elderly or disabled individuals in living independently, including fall detection, medication reminders, and emergency response systems.
- **Pet Care:** Automated pet feeders, cameras, and temperature control can help pet owners take care of their animals while away from home.
- **Fire Detection:** Sensors can detect smoke and ring alarm along with ensuring safety of the occupants by cutting off the electricity and automatically switching on the water hoses for extinguishing the fire. It can also detect the location of fire and suggest ways the fire can be extinguished, if it can be controlled.
- **Water Leak Detection:** Sensors can detect water leaks and automatically shut off the water supply to prevent damage.

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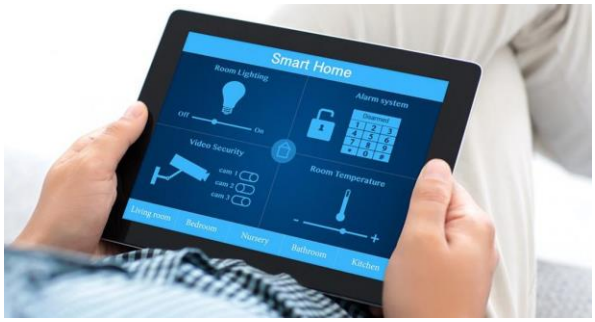
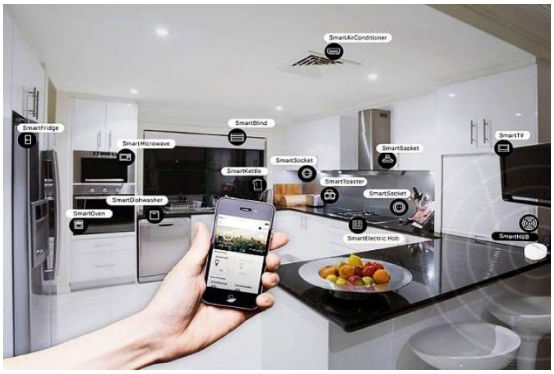

Smart Home
Room Lighting
Off
On
Alarm system
Video Security
cam 1
cam 2
cam 3
Room Temperature
Living room
Bedroom
Nursery
Bathroom
Kitchen

AVENGERS


SmartAirConditioner
SmartFridge
SmartBlind
SmartMicrowave
SmartSocket
SmartSocket
SmartTV
SmartKettle
SmartToaster
SmartSocket
SmartDishwasher
SmartOven
SmartElectric Hob
SmartHub

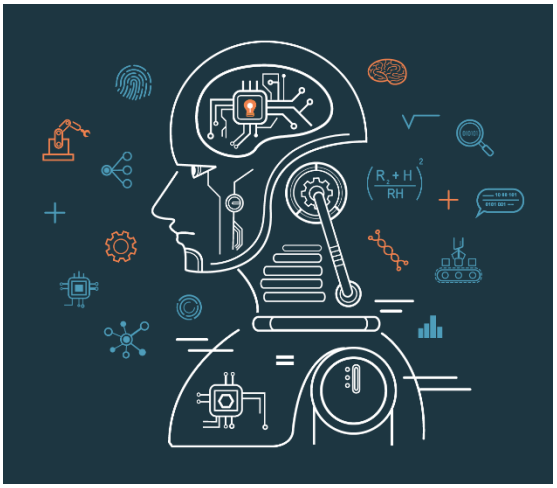
Future Scope

The future of home automation holds exciting prospects, as technology continues to advance and consumer demand for smart and connected homes grows. The future of home automation will be characterized by increased connectivity, AI-driven automation, enhanced user experiences, and a continued focus on sustainability and security. As technology continues to advance, the possibilities for improving the way we live and interact with our homes will only expand. Here are some key trends and future possibilities in the scope of home automation:

- **Increased Integration:** Future home automation systems will become even more integrated, allowing for seamless control of multiple devices and systems through a single interface or voice commands. This will simplify user experiences and make it easier to manage various aspects of the home.
- **Artificial Intelligence (AI) and Machine Learning:** AI will play a significant role in home automation. Smart systems will learn user preferences, adapt to changing conditions, and anticipate needs. For example, AI can optimize energy usage, suggest automation routines, or provide personalized recommendations for entertainment and comfort.
- **Energy Efficiency:** Home automation will continue to focus on energy conservation. Advanced algorithms and sensors will help homeowners reduce energy consumption by optimizing heating, cooling, lighting, and appliance usage. Renewable energy integration and grid interaction may also become more prevalent.
- **Health and Wellness:** The integration of health-monitoring devices and sensors in smart homes will enable individuals to track their well-being more effectively. These systems may provide real-time health data, medication reminders, and alerts for medical emergencies.
- **Security and Privacy:** With the increasing number of connected devices, security and privacy concerns will become more critical. Future home automation systems will need to incorporate robust

security measures, including encryption, secure authentication, and regular software updates to protect against cyber threats.

- **Network and Connectivity:** The rollout of 5G networks will enable faster and more reliable connections for smart home devices. This will facilitate real-time control, reduce latency, and support more bandwidth-intensive applications.
- **Augmented Reality (AR) and Virtual Reality (VR):** AR and VR technologies could be integrated into smart home systems for immersive experiences, such as virtual tours of homes, interactive interior design, or gaming within a smart environment.
- **Environmental Sustainability:** Smart homes of the future may prioritize sustainability and eco-friendliness. This includes features like rainwater harvesting, green roofs, and automated waste recycling systems.
- **Smart Cities Integration:** As smart city initiatives grow, home automation systems may become more closely integrated with municipal services, transportation networks, and public infrastructure to create a seamless urban living experience.
- **Human-Robot Interaction:** Robots and robotic assistants may become more common in smart homes, providing services like cleaning, security, and companionship.



References

- https://www.ibm.com/topics/industry-4-0#:~:text=and%20share%20data.-,Fourth%20industrial%20revolution,productively%20across%20the%20value%20chain.
- https://aws.amazon.com/what-is/iot/#:~:text=The%20term%20IoT%2C%20or%20Internet,as%20between%20the%20devices%20themselves.
- https://projecthub.arduino.cc/Shubhamkumar97/home-automation-using-arduino-and-bluetooth-control-2fd190
- https://en.wikipedia.org/wiki/Home_automation
- https://www.safewise.com/home-security-faq/how-does-home-automation-work/